

Test for the mean. Population variance known

Data scientist salary

Background You are given the data from the lesson
Task 1 Test the null hypothesis at 10% significance

Solution:

Dataset

\$ 117,313
\$ 104,002
\$ 113,038
\$ 101,936
\$ 84,560
\$ 113,136
\$ 80,740
\$ 100,536
\$ 105,052
\$ 87,201
\$ 91,986
\$ 94,868
\$ 90,745
\$ 102,848
\$ 85,927
\$ 112,276
\$ 108,637
\$ 96,818
\$ 92,307
\$ 114,564
\$ 109,714
\$ 108,833
\$ 115,295
\$ 89,279
\$ 81,720
\$ 89,344
\$ 114,426
\$ 90,410
\$ 95,118
\$ 113,382

Sample mean \$ 100,200
Population std \$ 15,000
Standard error \$ 2,739

Sample size 30

Z-score -4.67

Task 1 *Two-sided test*
 $z_{0.05}$ 1.65

4.67 > 1.65, therefore, we reject the null hypothesis

There was no need for the test as we have already rejected it lower significance levels. You could simply say that you would reject the null hypothesis.

- z = the z -score
- x = the specific data value you are looking at
- μ (or $\bar{x}$) = the mean (the average value of all data points)
- σ (or s) = the standard deviation (a measure of how spread out the numbers are)

H_0 (Glassdoor data) \$ 113,000
Two-sided test

- A **positive z -score** means the value is **above** the average.
- A **negative z -score** means the value is **below** the average.
- A **z -score of 0** means the value is exactly **equal** to the average.

For a **population**:

$$z = \frac{x - \mu}{\sigma}$$

For a **sample**:

$$z = \frac{x - \bar{x}}{s}$$

